

WASP-11b

R-0893

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_180121 · created 2026-07-29T02:34:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458139.6961 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.05
Transit depth (Rp/Rs)^2	3.02 +/- 1.74 [%]
Orbital Inclination (inc)	89.02 +/- 1.7
Airmass coefficient 1 (a1)	0.52 +/- 0.88
Airmass coefficient 2 (a2)	1.99 +/- 1.64
Scatter in the residuals of the lightcurve fit is	1121.13 %
Best Comparison Star	#1 - [459, 210]
Optimal Aperture	2.98
Optimal Annulus	7.71
Transit Duration (day)	0.104 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.05 vs 0.131 (deviation 0.41σ)
Duration fit vs published	0.104 d vs 0.1075 d (deviation 0.12σ)
Mid-time O–C	-3.4 min
Depth SNR	1.7
Residual scatter	1121.13 %

Light curve

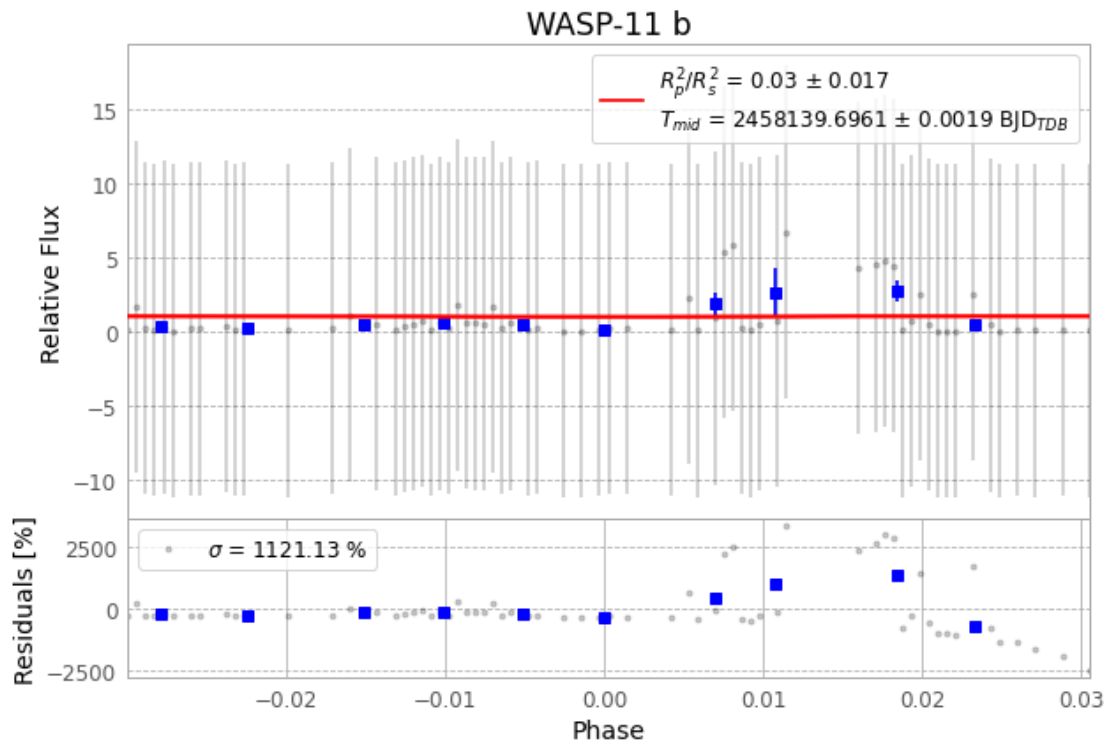


Figure 1 — FinalLightCurve_WASP-11 b_2018-01-20.png (SHA-256 a14c2cf9e7c02f3a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [574, 420], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2e4b347840e44ff5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

109 FITS frames (72 MB), HATP-10180121015712.FITS → HATP-10180121072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-180121.log (1615621f204c...), FinalLightCurve_WASP-11 b_2018-01-20.png (a14c2cf9e7c0...), FinalLightCurve_WASP-11 b_2018-01-20.csv (e8dc2be4fd73...)

Compute resources

Wall time 275.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 02:34Z.